

Full Carlitz Torsion Dynamics and Ramified Prime-Power Towers

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Abstract

We study the full Carlitz ring action over $\mathbb{F}_q(\theta)$, for every prime power q , rather than one additive polynomial in fixed characteristic. Every conductor torsion module is cyclic, and its annihilator strata give the exact preperiod and eventual period of every multiplier. Fixed-point counts and finite determinants follow without treating nonunit multipliers as permutations. An annihilator argument proves the specific reduction $C_P(X) \bmod P = X^{q^{\deg P}}$. Primitive polynomials at every prime-power level are Eisenstein. Their roots generate the full unit-group Galois action; finite-prime ramification proves linear disjointness and the corresponding theorem at every conductor. The source theorem is a synthesis of classical Carlitz cyclotomic theory, not a literature-priority claim or a target Riemann spectral construction.

Keywords: Carlitz module; torsion dynamics; annihilator strata; prime-power towers; ramification; source determinant.

中文摘要

本文研究一般特征下的完整 Carlitz 环作用，以全部导子扭点模为相空间，给出每个乘子作用的精确前周期、最终周期与原生有限行列式。关键约化由消去理想与不同根数矛盾证明，不把任意加性多项式误当 Frobenius。所有素多项式幂层的原始多项式均满足 Eisenstein 判据，进而得到完整 Galois 单位群、不同素导子塔的线性无交与任意导子结论。这些结果属于经典源系统理论的自包含整合，不宣称文献首创，也不提供目标零点或量子构造。

关键词：Carlitz 模；扭点动力学；消去理想分层；素幂塔；分歧；原生行列式。

1 Frozen source and the complete torsion dynamics

Put $A = \mathbb{F}_q[\theta]$, $K = \mathbb{F}_q(\theta)$, and $C_\theta(X) = \theta X + X^q$. The base field is exactly K . For $a = \sum a_j \theta^j$, define $C_a = \sum a_j C_\theta^{\circ j}$. Constants from $\mathbb{F}_q$ commute with C_θ , so $C_{a+b} = C_a + C_b$ and $C_{ab} = C_a \circ C_b = C_b \circ C_a$. For monic nonzero a , induction gives $\deg_X C_a = q^{\deg a}$ and $C'_a = a \neq 0$. Thus $\Lambda_a = \ker C_a \subset K^{\text{sep}}$ has exactly $q^{\deg a}$ distinct points. Generic characteristic refers to the injective coefficient map $A \rightarrow K$; the field still has characteristic p when $q = p^e$. Reduced torsion means multiplicity one, not Lyapunov stability.

The classical ring and cyclotomic framework is discussed in Conrad's author exposition [1] and in the tower background of Anglès–Bandini–Bars–Longhi [2]. The following proofs do not use their Iwasawa main conjecture. Our contribution within this repository is an all-conductor ring-action and ramification package: C384 studied a single map $x + x^p$ and its wild geometric zeta, while general finite-map determinants were already available in C204. Those finite formulas alone are not credited as the new mechanism.

Round 0 advance. We close the full annihilator/clock classification and the critical prime-reduction lemma, including nonunit and zero multipliers.

Theorem 1 (All conductor modules and multiplier clocks). *For every monic nonzero a , $\Lambda_a \cong A/a$ as an A -module. Points of exact annihilator $D \mid a$ have population $\Phi(D) = |(A/D)^\times|$, with $\Phi(1) = 1$. For $b \neq 0$, define*

$$D_* = \prod_{P \nmid b} P^{v_P(D)}, \quad h_D = \max_{P \nmid b} \left\lceil \frac{v_P(D)}{v_P(b)} \right\rceil.$$

The empty maximum is zero. Under C_b , a point in this stratum has exact preperiod h_D and eventual period $\text{ord}_{D_*}(b)$, with $\text{ord}_1(b) = 1$. For $b = 0$, every nonzero point reaches the fixed point zero in one step.

Proof. For P irreducible and $k \geq 1$, put $Q = q^{\deg P}$. There are Q^k points in Λ_{P^k} but only Q^{k-1} in $\Lambda_{P^{k-1}}$. Choose λ_k in their difference. Its annihilator ideal contains (P^k) but not P^{k-1} , hence is (P^k) . The injection $A/P^k \rightarrow \Lambda_{P^k}$ sending r to $C_r(\lambda_k)$ is bijective by cardinality. If u, v are coprime and $ru + sv = 1$, then $x = C_{sv}(x) + C_{ru}(x)$ splits Λ_{uv} into its u and v torsion, with zero intersection by Bezout. Induction proves the CRT and cyclicity. Exact-annihilator- D points are the classes $(a/D)r$ with $r \in (A/D)^\times$, giving their stated count.

The annihilator after one step is $D/\gcd(D, b)$, because $C_c(C_b x) = 0$ if and only if $D \mid cb$. After j steps the exponent at $P \mid b$ is $\max(v_P(D) - jv_P(b), 0)$, whereas other exponents are unchanged. A strictly decreasing annihilator cannot occur on a cycle. Exactly at h_D the remaining annihilator is D_* , where b is a unit. Return after n steps is then equivalent to $D_* \mid b^n - 1$. The assertions for $b = 0$, for $b = 1$, and for the one-point module $a = 1$ follow directly from $C_0 = 0$ and $C_1 = X$. $\square$

Proposition 1 (Finite source counts and determinant). *For $n \geq 1$ and every $b \in A$,*

$$F_n(a, b) = \# \text{Fix}(C_b^n | \Lambda_a) = q^{\deg \gcd(a, b^n - 1)}, \quad O_n = \frac{1}{n} \sum_{d|n} \mu(n/d) F_d.$$

For pullback U_b on all complex functions on Λ_a ,

$$\det(I - zU_b) = \prod_{d \geq 1} (1 - z^d)^{O_d}, \quad \exp \left(\sum_{n \geq 1} F_n z^n / n \right) = \det(I - zU_b)^{-1}.$$

Proof. The kernel of multiplication by $c = b^n - 1$ on A/a consists of the multiples of $a/\gcd(a, c)$, giving its size. Set $\gcd(a, 0) = a$. Every finite functional graph is a disjoint union of directed cycles with incoming trees, so $F_n = \sum_{d|n} dO_d$ and Möbius inversion gives integer nonnegative O_d . Ordering vertices by distance to a cycle makes the point-map matrix block triangular: the transient block is nilpotent, and a length- d cycle contributes $1 - z^d$. Transposing gives the pullback formula; expansion of $-\log(1 - z^d)$ proves the formal zeta identity. $\square$

In particular, a nonunit multiplier is not generally a permutation of the whole torsion module. The two clocks here are multiplier iteration n and conductor depth k ; neither is declared to be a rational-prime logarithm.

2 The critical Carlitz reduction

Lemma 1. *For every monic irreducible P , reduction of coefficients modulo P gives $\overline{C}_P(X) = X^Q$, where $Q = q^{\deg P}$.*

Proof. The reduced polynomial is monic of degree Q and has derivative zero. Suppose it has a nonzero root γ over $\overline{\mathbb{F}_Q}$. The map $a \mapsto \overline{C}_a(\gamma)$ has a proper annihilator ideal: it contains (P) , while $C_1(\gamma) = \gamma \neq 0$. Maximality of (P) identifies this annihilator as exactly (P) , so the ring orbit has Q distinct elements. Commutation makes them all roots of $\overline{C}_P$. But a degree- Q polynomial with derivative zero in characteristic p has every root of multiplicity at least p , and cannot have Q distinct roots. Thus zero is the only root; monicity gives X^Q . $\square$

This proves a specific ring-action property. It is not an assumption that an arbitrary additive polynomial reduces to Frobenius.

3 All prime-power Galois towers and composite conductors

Round 1 advance. We add primitive irreducibility, full Galois surjectivity, valuation ownership, and a finite-prime proof of disjointness at arbitrary conductors.

Theorem 2. *For $k \geq 1$, let $e_k = (Q - 1)Q^{k-1}$ and*

$$\Psi_P = C_P/X, \quad \Psi_{P^k} = \Psi_P \circ C_{P^{k-1}}.$$

This is a monic P -Eisenstein polynomial of degree e_k whose roots are exactly the primitive P^k -torsion points. For any such root λ_k ,

$$L_{P^k} = K(\lambda_k), \quad \text{Gal}(L_{P^k}/K) = (A/P^k)^\times, \quad \sigma_u(\lambda_k) = C_u(\lambda_k).$$

The prime P is totally ramified, $v_P(\lambda_k) = 1/e_k$, and every other finite prime is unramified. For all monic a , $\text{Gal}(L_a/K) = (A/a)^\times$.

Proof. The identity $C_P(Y) = Y\Psi_P(Y)$ gives $C_{P^k} = C_{P^{k-1}}\Psi_{P^k}$. All roots of the left side are simple, so the quotient has precisely the primitive roots. Reduction gives $\overline{\Psi}_{P^k} = X^{e_k}$, while its constant coefficient is exactly P . Thus every nonleading coefficient is divisible by P , but the constant coefficient is not divisible by P^2 . Eisenstein proves irreducibility. A primitive root generates the entire torsion module, so its field is the splitting field. Galois acts faithfully by units; the irreducible degree equals the number of units, proving surjectivity.

At a place over P , an Eisenstein root has positive valuation. Otherwise the monic term is uniquely minimal. With positive valuation every nonconstant, nonleading term has valuation greater than one; cancellation forces $e_k v_P(\lambda_k) = 1$. Its denominator forces ramification index at least e_k , and the global degree bounds it above. Hence the prime has one place, residue degree one, and total ramification.

At a finite $R \neq P$, the monic polynomial C_{P^k} has derivative P^k , a unit. Its residue roots are distinct in a finite residue extension. Pass to the associated unramified extension of the complete field K_R . Hensel lifting gives all the roots there, so the splitting field is unramified at R .

Now form a compositum of distinct-prime towers inductively. At the next prime P , the previous compositum is unramified, while the new tower is totally ramified. Their intersection is both. A subextension of a globally totally ramified extension has one place with local degree equal to global degree; an extension both total and unramified has degree one. Hence the intersection is K . The two Galois extensions are linearly disjoint. Multiplying degrees and using the module CRT identifies the compositum with L_a and gives exactly $(A/a)^\times$. $\square$

Infinity ramification and the full maximal abelian extension are outside this theorem; constant extensions are not silently added.

4 Conclusion

The full ring action, rather than an isolated additive-map census, makes annihilator ideals and coefficient primes intrinsic to the source. Primitive Eisenstein polynomials close the full finite Galois and composite-conductor construction. These classical source mechanisms are not a target Riemann construction. The audit records exact finite regressions separately from the all-parameter proof; no finite evidence or formal determinant upgrades the target claim.

References

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